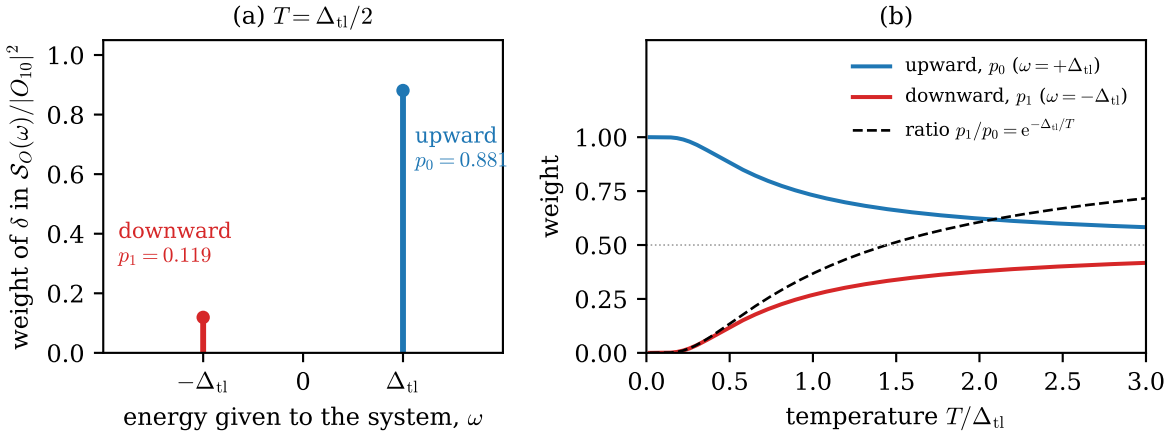
(a) $T = \Delta_{\mathrm{tl}}/2$
weight of δ in $\mathcal{S}_O(\omega)/|O_{10}|^2$
1.0
0.8
0.6
0.4
0.2
0.0
downward
$p_1 = 0.119$
upward
$p_0 = 0.881$
$-\Delta_{\mathrm{tl}}$
0
Δ_{tl}
energy given to the system, ω
(b)
upward, p_0 ($\omega = +\Delta_{\mathrm{tl}}$)
downward, p_1 ($\omega = -\Delta_{\mathrm{tl}}$)
ratio $p_1/p_0 = \mathrm{e}^{-\Delta_{\mathrm{tl}}/T}$
weight
1.00
0.75
0.50
0.25
0.00
0.0
0.5
1.0
1.5
2.0
2.5
3.0
temperature T/Δ_{tl}